EXPERIMENT 10

Aim: To demonstrate Distributed File System

Objective:- Demonstrate Distributed File System using HDFS

Theory:-

HDFS employs a NameNode and DataNode architecture to implement a distributed file system that provides high-performance access to data across highly scalable Hadoop clusters.

File management tasks in hadoop

In order to perform operations on Hadoop like copy, delete, move etc., following steps can be used:

Basic operations:

1. Create a directory in HDFS at given path(s).

Usage:

hadoop fs -mkdir <paths>

2. List the contents of a directory.

Usage :

hadoop fs -ls <args>

3. See contents of a file

Same as unix cat command:

Usage:

hadoop fs -cat <path[filename]>

4. Copy a file from source to destination

This command allows multiple sources as well in which case the destination must be a directory.

Usage:

hadoop fs -cp <source> <dest>

5. Copy a file from/To Local file system to HDFS

copyFromLocal

Usage:

hadoop fs -copyFromLocal <localsrc> URI

Similar to put command, except that the source is restricted to a local file reference.

copyToLocal

Usage:

hadoop fs -copyToLocal [-ignorecrc] [-crc] URI <localdst>

Similar to get command, except that the destination is restricted to a local file reference.

7. Move file from source to destination.

Note:- Moving files across filesystem is not permitted.

Usage :

hadoop fs -mv <src> <dest>

8. Remove a file or directory in HDFS.

Remove files specified as argument. Deletes directory only when it is empty

Usage :

hadoop fs -rm <arg>

Steps for copying file

1) Go to Hadoop folder and then to sbin

C:\>cd C:\hadoop-2.8.0\sbin

2) Start namenode and datanode with this command, Two more cmd windows will open

C:\hadoop-2.8.0\sbin>start-dfs.cmd

3) Now start yarn through following command, Two more windows will open, one for yarn resource manager and one for yarn node manager

C:\hadoop-2.8.0\sbin>start-yarn.cmd

4) Create a directory named 'sample' in the hadoop directory using the following command

C:\hadoop-2.8.0\sbin> hdfs dfs -mkdir /sample

5) To verify if the directory is created

C:\hadoop-2.8.0\sbin>hdfs dfs -ls /

6) Copy text file from D drive to sample

C:\hadoop-2.8.0\sbin>hdfs dfs -copyFromLocal d:\rally.txt /sample

7) To verify if the file is copied

C:\hadoop-2.8.0\sbin>hdfs dfs -ls /sample

Code and output:

1. Create a Directory in HDFS

Command:

```
hadoop fs -ls /user/hadoop/
```

Output: (No output if successful, directory gets created)

Verify using:

```
hadoop fs -ls /user/hadoop/
```

Expected Output:

```
Found 1 items
drwxr-xr-x   -  hadoop  supergroup          0  2025-03-17  12:00
/user/hadoop/mydata
```

2. List Contents of a Directory

Command:

```
hadoop fs -ls /user/hadoop/
```

Output:

```
Found 2 items
drwxr-xr-x   -  hadoop  supergroup          0  2025-03-17  12:00
/user/hadoop/input
drwxr-xr-x   -  hadoop  supergroup          0  2025-03-17  12:05
```

```
/user/hadoop/output
```

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3. View Contents of a File in HDFS

Command:

```
hadoop fs -cat /user/hadoop/mydata/sample.txt
```

Output:

```
Hello, this is a sample file stored in HDFS.
```

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4. Copy a File from One HDFS Location to Another

Command:

```
hadoop fs -cp /user/hadoop/mydata/sample.txt /user/hadoop/backup/sample_copy.txt
```

Verify using:

```
hadoop fs -ls /user/hadoop/backup/
```

Output:

```
-rw-r--r--   3 hadoop supergroup   40 2025-03-17 12:10 /user/hadoop/backup/sample_copy.txt
```

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5. Copy a File from Local to HDFS

Command:

```
hadoop fs -copyFromLocal /home/user/data.txt /user/hadoop/mydata/
```

Verify using:

```
hadoop fs -ls /user/hadoop/mydata/
```

Output:

```
-rw-r--r--   3 hadoop supergroup   40 2025-03-17 12:15 /user/hadoop/mydata/data.txt
```

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6. Copy a File from HDFS to Local

Command:

```
hadoop fs -copyToLocal /user/hadoop/mydata/sample.txt /home/user/
```

Verify using:

```
ls -l /home/user/
```

Output:

```
-rw-r--r--   1 user  user  40 2025-03-17 12:20 /home/user/sample.txt
```

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7. Move a File in HDFS

Command:

```
hadoop fs -mv /user/hadoop/mydata/sample.txt /user/hadoop/archive/
```

Verify using:

```
hadoop fs -ls /user/hadoop/archive/
```

Output:

```
-rw-r--r--   3 hadoop supergroup   40 2025-03-17 12:25 /user/hadoop/archive/sample.txt
```

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8. Remove a File in HDFS

Command:

```
hadoop fs -rm /user/hadoop/mydata/data.txt
```

Output:

```
Deleted /user/hadoop/mydata/data.txt
```

Conclusion:
HDFS is optimized for large-scale, batch processing in distributed environments, unlike traditional file systems (NTFS, ext4) that lack fault tolerance. Compared to cloud storage (S3, GCS), HDFS is better for on-premise Big Data processing, while cloud solutions excel in scalability and remote access. It’s widely used in data lakes, ETL, and ML workloads.